Recommended Equipment Stack for USA Battery Recycling Plant

Hybrid Approach: Chinese Equipment + Western Safety Systems

Target Capacity: 1-2 t/h (1,000-2,000 kg/h)
Estimated CAPEX: 500K– 1M (excluding installation)
Payback Period: 24-30 months

Equipment Recommendations

Stage	Equipment	Supplier	Cost	Notes
1. Primary Shredder	Double-shaft shear, nitrogen-ready	**Genox** (China) or **Xingmao** (China)	$25K-45K	Verified USA installations; Genox has US office
2. Hammer Mill	High-speed impact crusher, 1500 rpm	**Xingmao** or **Henan Doing** (China)	$15K-25K	1500 rpm optimal per MDPI research
3. Screening	Multi-deck vibrating screen (2mm, 5mm, 20mm)	**Xingmao** (integrated)	$8K-15K	Precision cut points critical for purity
4. Magnetic Separator	Rare earth drum magnet	**Eriez** (USA)	$10K-20K	Western required - industry standard
5. Eddy Current Separator	Non-ferrous metal separator	**Eriez** (USA)	$20K-35K	Higher recovery = faster payback
6. Air Classifier	Density separation	**Xingmao** (integrated)	$10K-20K	Chinese adequate
7. Dust Collection	HEPA filtration system	**Xingmao** + Western filters	$10K-18K	Use Western HEPA filters
8. Nitrogen System	Membrane generator + O2 monitoring	**Generon** (USA)	$30K-50K	**SAFETY CRITICAL**

Stage	Equipment	Supplier	Cost	Notes
9. Fire Suppression	Water mist + CO2 backup	**Siemens** (USA)	$25K-40K	**SAFETY CRITICAL**
10. Gas Detection	HF, HCN, CO sensors	**Drager** or **Honeywell** (USA)	$15K-30K	**SAFETY CRITICAL**

Total Estimated Cost

Category	Cost
Chinese Equipment (1-7)	$100K-180K
Western Safety + Separation (8-10)	$80K-135K
Equipment Total	**$180K-315K**
Installation, Integration, Commissioning	$80K-150K
Project Total	**$300K-500K**

Note: Lower end = 500 kg/h, higher end = 1-2 t/h

Why This Mix Works

- **Chinese equipment (60-70% savings):** Shredders, crushers, screens, and air classifiers are commodity equipment with mature global manufacturing
- **Western safety (non-negotiable):** Nitrogen systems, fire suppression, and gas detection require proven reliability for insurance approval
- **Western separation (worth premium):** Eriez separators deliver measurably better recovery rates - premium recovered in 12-18 months

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Top Supplier Contacts

Supplier	Product	Website
Genox (China)	Complete lines, shredders	genoxtech.com
Xingmao (China)	Complete lines, verified USA installs	lithiumbatterybroken.com
Eriez (USA)	Magnetic & eddy current separators	eriez.com

Supplier	Product	Website
Generon (USA)	Nitrogen generators	generon.com
Drager (USA/Germany)	Gas detection systems	drager.com
Siemens (USA)	Fire suppression	siemens.com

Critical Success Factors

1. **Feedstock first** - Secure 2-3 year battery supply agreements before purchasing
2. **Offtake contracts** - Identify black mass buyers (Li-Cycle, Redwood, etc.)
3. **Safety budget** - Do not reduce Western safety system allocation
4. **Installation planning** - Budget 25-35% above equipment cost for installation
5. **Spare parts** - Order 6-month spare parts inventory with equipment delivery